

13th Feb 2026 / CSC370

⇒ Translaⁿ Problem

(Tamil → English)

- Decision lists - simple rule-based classificaⁿ in ML. A decision list is an ordered list of if-then rules.
- When you have map, then you don't have to learn it, then there is no req. of bottom's-up approach to solve that problem → just a generative model.
- piece vs pease → identifying candidates with homophones (they ^{not works} sound the same)

which we have to learn

Estimate

$$P(w_i | c_{-k} \dots c_k) \propto$$

$$\frac{P(c_{-k} \dots c_k | w_i)}{P(c_{-k} \dots c_k)} = \frac{P(w_i | c_{-k} \dots c_k)}{P(w_i)}$$

one set of params ← other set ← priors

2 types of context

- context words → surrounding words around a target word in a sentence
- collocations → words that frequently occur together in a lang.

- until we see the w_i , the context words around w_i , cannot be considered as context.
- Brown corpus → a structured collecⁿ of eng texts from diff. sources, used for lang. analysis.

- _/_/_
- hard problem of calculating priors & likelihoods, also problem of identifying the candidates.
 - currently we have the assumption all the words around us are all actually having correct spellings. Otherwise, this 'll be nasty circular problems.
 - Circularity (interdependency) is crucial for our problem.
 - difference btw. test-train split & cross-validation
→ for evaluation
 ↳ train data, used to train the params
 ↳ to make sure your train data does not just overfit the model.
 - Thesis → Lang. is not self-sufficient, lang. is just a way to fulfill partial things coming from your heart.
 - Perceptual grounding problem of AI - challenge of connected words, symbols or concepts inside a comp. to real-world sensory experiences.
 - More you know, more you become silent abt. the same.